NAME ____________ DATE ____________

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1 COMP — Membrane composition and fluidity

You should be able to: Describe the fluid mosaic membrane and state how phospholipids, cholesterol, glycolipids, and integral and peripheral proteins contribute to its properties.

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2 COMP — Determinants of permeability

You should be able to: Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein.

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3

COMP

Simple diffusion and Fick's law

You should be able to: State the variables in Fick's law of diffusion and predict how a change in concentration gradient, surface area, membrane thickness, or distance changes the rate of transfer.

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4

COMP

Osmolarity and tonicity

You should be able to: Calculate the osmolarity of a solution, distinguish osmolarity from tonicity, and classify a solution as isotonic, hypotonic, or hypertonic to a cell.

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5 COMP — Osmosis and cell volume

You should be able to: Predict the direction of water movement and the resulting change in cell volume when a cell is placed in a solution of stated osmolarity and penetrating solute content.

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6 COMP — Diffusion and osmosis experiment

You should be able to: Measure diffusion and osmotic movement across a selectively permeable membrane and relate the observed rate to molecular size and concentration gradient.

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7 COMP Tonicity and red blood cells

You should be able to: Observe erythrocytes in solutions of different tonicity, identify crenation, normal shape, and hemolysis, and explain each result.

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8 COMP Facilitated diffusion

You should be able to: Describe carrier and channel mediated diffusion and explain why carrier mediated transport shows saturation and specificity while simple diffusion does not.

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9
COMP

Primary active transport

You should be able to: Explain how the sodium potassium ATPase uses ATP to move three sodium out and two potassium in, and state the two gradients it maintains.

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10
COMP

Secondary active transport

You should be able to: Distinguish symport from antiport and trace how the sodium gradient powers glucose uptake by SGLT and calcium removal by the sodium calcium exchanger.

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11 COMP Transport maximum and saturation

You should be able to: Interpret a transport rate curve, identify the transport maximum, and apply the concept to renal glucose handling in hyperglycemia.

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12 COMP Vesicular transport

You should be able to: Compare phagocytosis, pinocytosis, receptor mediated endocytosis, and exocytosis by trigger, cargo, and energy requirement.

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13 Transepithelial transport

COMP

You should be able to: Trace glucose or sodium from lumen to blood across a polarized epithelium and identify which step occurs at the apical membrane and which at the basolateral membrane.

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14 Transport simulation

COMP

You should be able to: Use a membrane transport simulation to distinguish simple diffusion from facilitated diffusion and active transport by their response to gradient reversal and metabolic poison.

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